Lesson 8 Teacher's Key: Under Pressure

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Extending GameManager - the dedicated sprite introduced in Lesson 6 for win-tracking - with a second, independent counter for a lose condition, reusing the same broadcast/receive pattern and the escape branch already built in Lesson 7.

Level 1 - Full Worked Solution

- lives is a second variable living on GameManager, exactly parallel to enemies_remaining - same sprite, same broadcast-based announcement pattern, different broadcast name and different trigger condition.
- The Enemy's escape branch (y position < -170) now broadcasts enemy escaped instead of touching lives directly - the Enemy only ever reports what happened, it never counts or decides anything, exactly like the touching-Bullet branch already does for enemy hit.
- GameManager's new "when I receive [enemy escaped]" script owns the decrement, the zero-check, and the game-over broadcast - a direct parallel to its existing "when I receive [enemy hit]" script.
- "lives = 0" (exact equality) mirrors "enemies_remaining = 0" from Lesson 6 exactly - the same reasoning applies: a variable that only ever decreases by 1 can only cross zero once, so exact equality is safe and simpler than <=.

Highlighted lines are new this week; everything else is code you already have.

```
// GameManager (hidden sprite) - existing green flag script:
when green flag clicked
hide
set enemies_remaining to 5
set lives to 3

// Enemy sprite - the escape branch inside its forever loop:
if y position < -170 then
    broadcast [enemy escaped]
    delete this clone

// GameManager - new script:
when I receive [enemy escaped]
change lives by -1
if lives = 0 then
    broadcast [game over]

// Player sprite - new script:
```

```
when I receive [game over]
say [Game Over] for 3 seconds
```

Why This Structure

- This is a deliberate payoff for good Lesson 6/7 structure: because GameManager already owns all counting and decision-making, and the escape branch was already cleanly separated from the touching-Bullet branch, adding a second consequence required only one new broadcast and one new receiving script - zero risk of accidentally touching enemies_remaining.
- Using broadcast/receive again, with GameManager as the single listener, keeps this lesson consistent with Lesson 6's win-announcement pattern and Lesson 7's Rule 13 retrofit - students are extending a tool and a structure they already trust, not learning either one fresh.

Level 2 / Level 3 Talking Points

- Level 3 sound, worked (Scratch): import the provided sfx_gameover.wav and sfx_lifelost.wav. On the Player, "when I receive [game over]" becomes: start sound [sfx_gameover] -> say [Game Over] for 3 seconds. On GameManager, "when I receive [enemy escaped]" becomes: change lives by -1 -> start sound [sfx_lifelost] -> if lives = 0 then broadcast [game over]. Expect both sounds to overlap on the final life - acceptable, or students can wrap the lifelost sound in an if lives > 0.
- Ask students to point out exactly which broadcast the Enemy sends on escape, and which script on GameManager receives it - checking their own reasoning out loud catches the most common mistake (touching lives from the wrong place) before it happens.
- Level 3 visual: the built-in variable-watcher checkbox is worth calling out explicitly as "the Scratch equivalent of draw_text() or print()" - genuinely the simplest possible way to show a running number, no blocks needed at all.

Level 2 - complete code:

```
// Level 2 is verification only - the Level 1 code is already correct, so
no lines change.
// Checks: shooting an enemy never changes lives (only the kill path
changes enemies_remaining);
// an escape never changes enemies_remaining (only the escape path changes
lives);
// "Game Over" appears once, when lives first reaches 0.
```

Level 3 - complete code (all three goals combined):

Highlighted lines are the Level 3 additions on top of the Level 1 code.

```
// Player sprite - Level 3 sound:
when I receive [game over]
start sound [sfx_gameover]
say [Game Over] for 3 seconds

// GameManager - Level 3 sound:
when I receive [enemy escaped]
change lives by -1
start sound [sfx_lifelost]
if lives = 0 then
    broadcast [game over]
```

```
// Level 3 visual: tick the checkbox beside the lives variable in the
Variables palette.
// On the last life both sounds overlap - acceptable, or wrap start sound
[sfx_lifelost] in: if lives > 0.
```

Common Student Mistakes

- Adding a lives-changing block directly to the Enemy sprite instead of broadcasting enemy escaped and letting GameManager handle it - even if the number comes out right, this bypasses the structure the rest of the course relies on.
- Broadcasting enemy escaped from the touching-Bullet branch instead of the escape branch (makes shooting cost a life, which is backwards).
- Using "lives < 1" or forgetting the check entirely and just broadcasting every frame lives is low.